

Retrieval based Voice Conversion

What is it, and what is it not?

Retrieval based Voice Conversion (RVC) is NOT text-to-speech synthesis. It is a Speech-to-Speech AI voice synthesizer framework.

This is to say, you speak into it, and it spits back what you spoke. However it's in the “model's” voice, while retaining the voice acting.

Language is very powerful and complex. Where Text-to-Speech fails is ability to convey emotion via tone, pauses, emphasis, pitch, etc.

Where Speech-to-Speech excels is in retaining those qualities. However, output is dependant on training data and the voice actor's performance.

What do I need to run it?

Local Hosting

- Requires a CUDA compatible Nvidia GPU
 - AMD rCOM not currently supported, though its in the pipeline with Pytorch DirectML.
 - Inference can be done on CPU but its very slow.
- Platform agnostic
 - AMD rCOM is only supported on Linux.
- ~30GB+ of disk space recommended for

Cloud Hosting

- Google Colab
- Cloud hosted GPUs
 - DigitalOcean/Paper Space
 - Google
 - Others

My Setup

- Hypervisor is an HP DL380e Gen 8 running Proxmox
- GPU is an Nvidia Tesla P40, 24GB VRAM
 - Tesla K40 12GB alternative
 - Has no active cooling
 - Requires Resizable BAR and Above 4G Decoding Motherboard
- VM is running Ubuntu 22.04
- Microphone is a Yeti Snowball
- Audio Technica ATH-M50x headphones

Voice Inference

Target Audio Requirements

Target audio, the audio you're converting, has a few requirements. As with most AI software, garbage in, garbage out.

Pure audio recorded with a clear microphone is best. No background noise or artifacts should be present within your audio, as it will distort your conversion.

Ideally, you have no background noise or music. However, if you do, you can use UVR5 (which is built in) to separate vocals and instrumentals.

Nice quality speakers or headphones are recommended to listen for issues within your audio.

Inference Demo

Voice Models

Model Training

A model is basically the result of what audio you fed during training.

Model training has a lot of nuance, in a lot ways, it's a black box.

The most important aspect is quality, then quantity, then diversity of your training audio. Having audio which is clear is the most important. Having more of it helps. And having a diverse set, i.e. emotion or emphasis, is nice to have.

Depending on how much source material you have, how fast your hardware is, and how much 'training' you want to be put into the model determines how long it will take.

Model Training Pt. 2

Models are trained on an amount of 'epochs'. Think of these as units. There is a possibility to 'over' and 'under' train a model.

General Guidelines:

- ~10 minutes of isolated training data.
- ~300 epochs to train to
- A model will be about ~500mb to ~2GB

"I don't want to train models"

<https://huggingface.co/models?other=rvc>

I don't want to train models. Where can I find them?

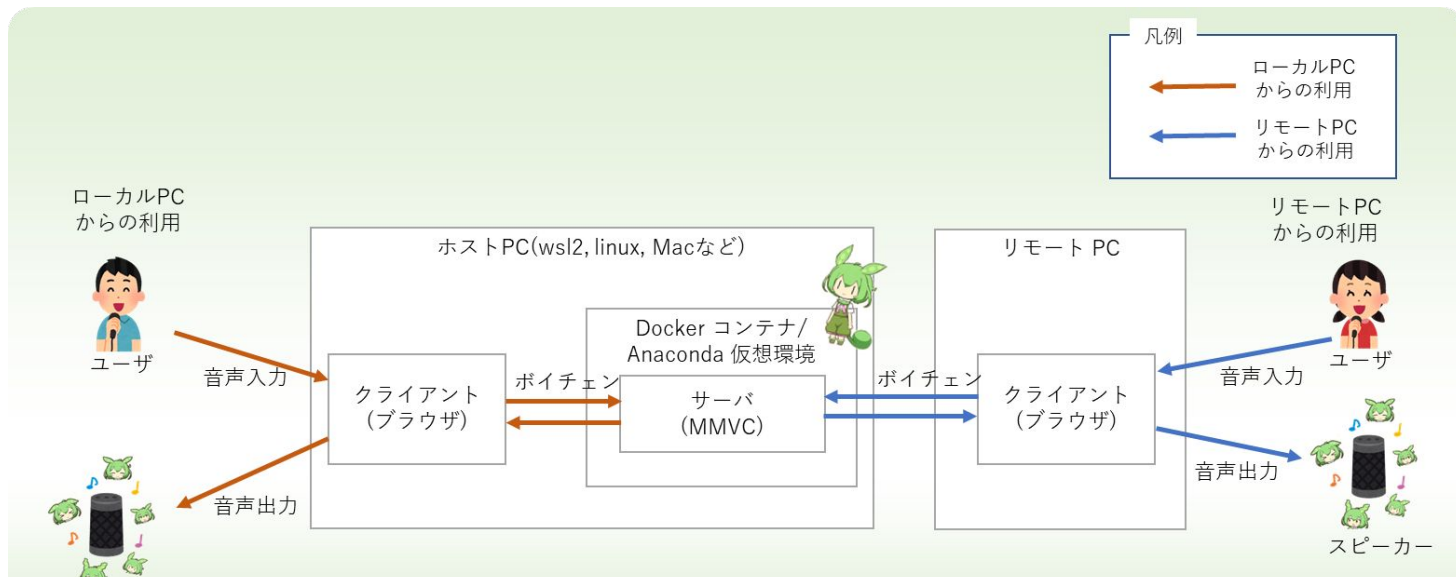
- Posted on Hugging Face
 - <https://huggingface.co/models?other=rvc>
- Discord Servers
 - <https://discord.com/invite/aihub>
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Training Demo

Realtime Voice Changer

External project which can be used as a server or local service to change your voice in realtime by feeding it into RVC. View links for setup instruction.

Image depicts remote and local hosting and realtime conversion.



Use Cases and Viability

- Memes
- Independent content creation
 - Video games, animation, etc.
 - Modding
 - Audiobooks
- Impersonation
 - Providing anonymity online
 - Separate online identities
 - Vtubers
 - Can be treated as filters, but for your voice
- Malicious Intent

Importance of Self Hosting and Open Source AI

Nobody can take away your source or your hardware. Nobody can tell you what to do with it. It is yours.

Many view recent developments within the AI sphere to be the dawn of our next industrial/information revolution, and who controls this has not been defined

AI safety & ethics has been a hot button issue, which is being used to justify calls to regulate and censor AI technologies.

Google's leaked "We Have No Moat" internal memo, discussing how they viewed the open source community as their greatest competition.

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RBVC GitHub:

<https://github.com/RVC-Project/Retrieval-based-Voice-Conversion-WebUI>

Real Time Voice Conversion: <https://github.com/w-okada/voice-changer>

Try It with Colab:

https://colab.research.google.com/github/RVC-Project/Retrieval-based-Voice-Conversion-WebUI/blob/main/Retrieval_based_Voice_Conversion_WebUI.ipynb

Models on HuggingFace: <https://huggingface.co/models?other=rvc>